

Exercise Deck 1

Option Pricing

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Instructions:

Show all your work. Justify all steps clearly. Collaboration is allowed but write your own solutions.

Programming assignments should be completed on Google Colab. Provide links to complete notebooks. One notebook per assignment. Structure and comment your code professionally. Annotate with clean markups. Use latex where appropriate.

Exercise solutions should be typed, preferable with latex. Provide complete pdf files, with your name and student number.

1. Exercise: Model-Free Arbitrage

Use hard arbitrage arguments, do not assume a model of any kind. Markets are transparent and frictionless. Rates and dividends are 0.

(a) A basket option delivers at maturity T the payoff:

$$\left(\sum_{i=1}^n \gamma_i S_T^i - K\right)^+$$

based on n underlying assets S^i , and weights γ_i summing to 1. Show that its price is less than (or equal to) the basket of options $\sum_i \gamma_i C^i(K, T)$ where $C^i(K, T)$ is the price of a call on asset i , strike K and maturity T .

(b) An Asian option is an option on the average underlying price over time:

$$\left(\frac{1}{n} \sum_{i=1}^n S_{T_i} - K\right)^+$$

where $T_1 < T_2 < \dots < T_{n-1} < T_n = T$ are fixing dates. Show that its price is less than (or equal to) the European call of strike K and maturity T .

2. Exercise: Large Drift

Assume zero rates. Assume that the underlying prices series of some (non-dividend bearing) asset follows the lognormal SDE:

$$\frac{dS_t}{S_t} = \mu dt + \sigma dW$$

with an incredibly large drift of $\mu = 150\%$ per annum, and volatility $\sigma = 5\%$ per annum. (This is a dream investment with sharp ratio 30, yet, not an arbitrage). Having learned in class that drift doesn't matter, you sell an ATM call for 6% Black and Scholes volatility and hedge daily. Assume for simplicity that a (business) day is $1/225 = 1/15^2$ years.

- (a) State the lognormal form of the mis-replication term taught in class, and show that it promises a daily profit.
 - (b) What is the actual distribution of $\frac{\Delta S}{S\sqrt{\Delta t}} \approx \frac{\Delta \log S}{\sqrt{\Delta t}}$?
 - (c) Express the daily break-even point. What is the probability of a positive PnL?
 - (d) Compute the expectation and variance of $\frac{PnL}{\Gamma S^2 \Delta t/2}$.
 - (e) Why are you bleeding money every day when the formula taught in class promises profit?
 - (f) Resolve the paradox, and explain what you can do to stop the bleeding and realize the daily profit promised by the theoretical formula. Explain how this demonstrates that drift, effectively, doesn't matter, but only if you implement a correct hedge strategy.
- [Hint: try hedging twice daily, then n times daily: what happens?]

3. Exercise: Hedging a mean-reverting process

Assume zero rates. Assume that the underlying prices series of some (non-dividend bearing) asset follows the mean-reverting SDE:

$$dS_t = -\kappa(S_t - S_0)dt + \sigma dW$$

- (a) Show that $S_T \sim N(S_0, \frac{1-e^{-2\kappa T}}{2\kappa}\sigma^2)$
- (b) Suppose that $S_0 = 100$, $\sigma = 10$ and $\kappa = 1$. Then one-year standard deviation is 6.56.

What is the correct price for a one-year option? Is it $Bach(10)$ or $Bach(6.56)$ or something else?

Explain why there is only one right answer, both in mathematical and trading terms, although the optimal hedging strategy depends on risk preference.

4. Exercise: Quadratic Variation Identity

Prove the arithmetic identity:

$$2 \sum_{i=0}^{n-1} (S_0 - S_i)(S_{i+1} - S_i) = \sum_{i=0}^{n-1} (S_{i+1} - S_i)^2 - (S_n - S_0)^2$$

5. Exercise: Bachelier Formula

Derive Bachelier's formula: assuming $dS_t = \hat{\sigma} dW_t^Q$,

$$E_t^Q[(S_T - K)^+] = \int_{\mathbb{R}} (x - K)^+ n(x; S_t, \sigma\sqrt{T-t}) dx = (S_t - K)N(d) + \hat{\sigma}\sqrt{T-t}n(d)$$

where $d = \frac{S_t - K}{\hat{\sigma}\sqrt{T-t}}$

6. Exercise: Greeks in Bachelier and Black and Scholes

Demonstrate the Greek formulas in Bachelier and Black and Scholes: delta, gamma, theta, vega.

Do not compute the partial derivatives of the pricing formulas directly, as this is tedious and unnecessary. Be smart, proceed in the right order, and use what you learned in class to derive these Greeks in a terse and elegant manner.

You may assume Black-Scholes $\Delta = N(d_1)$, since an elegant derivation requires Girsanov's theorem, and the derivation is otherwise tedious.

7. Exercise: Normal Variance Contracts**8. Exercise: Displaced Model**

The displaced-lognormal assumes an affine volatility of the form:

$$dS_t = (a + bS_t)dW_t^Q$$

so it generalizes Black and Scholes and Bachelier models. We need $b \neq 0$ in what follows, but it can be arbitrary small, so the model converges to Bachelier. Alternatively, it coincides with Black and Scholes when $a = 0$.

The displaced-lognormal model is a particular case of time-homogeneous local volatility model, where local volatility is restricted to linear functions. In return, the model offers great analytic tractability, as we will see.

- (a) What is the risk-neutral SDE of $X_t = a + bS_t$?
- (b) By expressing the payoff of a European call $(S_T - K)^+$ in terms of $(X_T - K)^+$, find the closed-form pricing formula for calls in the displaced-lognormal model.

9. Exercise: Lognormal Variance Contracts**10. Exercise: Mis-Replication Formula****11. Exercise: Greeks in Bachelier and Black and Scholes****12. Exercise: Time-Dependent Volatility****13. Exercise: Dupire's formula with implied volatility****14. Exercise: Harmonic approximation****15. Exercise: Digital options****16. Exercise: Butterflies****17. Exercise: General European Options****18. Exercise: General European Options****19. Exercise: Hedging in dS not dt****20. Exercise: Hedging with incorrect vol**